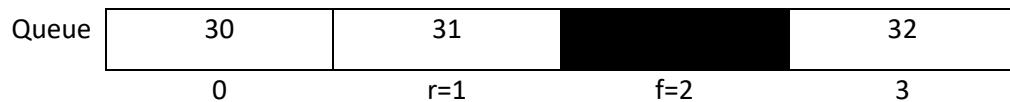


CSE 2215: Data Structure and Algorithms-I

- 1 a) What is STACK? What are the properties of STACK data structure?
- b) What are the applications of a STACK data structure?
- 2 Show the status of a STACK for the following operations assuming the STACK size, $m=2$, where the STACK is implemented by an array.
 Push(10), Push(20), Pop(), Push(30), Push(40), Pop(), Pop(), Pop()
- 3 Design an algorithm to reverse a string using a STACK
- 4 Implement a STACK using an array
- 5 Show the status of a STACK for the following operations using linked list
 Push(10), Push(20), Pop(), Push(30), Push(40), Pop(), Pop(), Pop()
- 6 Implement a STACK using linked list
- 7 Convert the following infix expression into postfix using a STACK.
 Infix expression: $a \uparrow 2 - (b + c/d * a + b)$
- 8 Evaluate the postfix expression, $a \ b - \ c \ d * +$ for $a=2$, $b=3$, $c=2$ and $d=1$ using a STACK
- 9 Design a recursive algorithm for TOWER OF HANOI using one intermediate pillar/peg and show simulation for $n = 4$, where n is the number of disks.
- 10 Design an iterative algorithm for TOWER OF HANOI using one intermediate pillar/peg and show simulation for $n = 4$, where n is the number of disks.
- 11 a) What is QUEUE? What are the properties of QUEUE data structure?
- b) What are the applications of a QUEUE data structure?
- 12 Show the status of a QUEUE for the following operations assuming the QUEUE size, $m=2$, where the QUEUE is implemented by an array.
 Insert(10), Insert(20), Delete(), Insert(30), Insert(40), Delete(), Insert(90) Delete(), Delete()
- 13 How is overflow checking in a QUEUE is done when it is implemented by an array?
- 14 Implement a QUEUE using an array
- 15 Show the status of a QUEUE using linked list.
 Insert(10), Insert(20), Delete(), Insert(30), Insert(40), Delete(), Insert(90) Delete(), Delete()
- 16 Implement a QUEUE using linked list
- 17 Show the mechanism of the following algorithm using the Queue of size 3. Here, Queue is a FIFO data structure, and m , f and r are size, front and rear of the Queue, respectively. What is the purpose of the algorithm?



```

i=(f+1)%(m+1);
while( i!=f ){
    printf("%d  ", Queue[i]);
    i=(i+1)%(m+1);
}

```

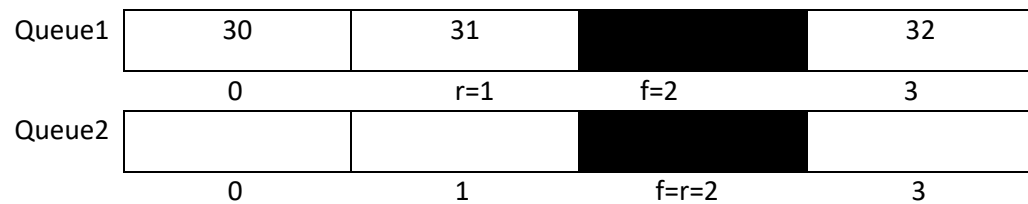
- 18 What are the differences and similarities between a QUEUE and a Priority QUEUE?
- 19 Show the status of a QUEUE and a Priority QUEUE for the following operations assuming the QUEUE size, $m=3$, where the QUEUE is implemented by an array.
 Insert(30), Insert(20), Delete(), Insert(25), Insert(40), Delete(), Insert(10) Delete(), Delete()

- 20 What are the applications of a Priority Queue?
- 21 Show the utilization of a Priority QUEUE in Heap Sort algorithm with an example.
- 22 Show the status of Queue1 and Queue2 for each of the iterations of the while loop given below

```

Count=1
Sum=0
while (Queue1 ≠ Empty)
    x ← Queue1
    Sum=Sum+x
    Queue2 ← Sum/Count
    Count=Count+1
End While

```



Discuss the Space Complexity of the following algorithms

1) <pre> int A[100], i, n, sum; float avg; scanf("%d", &n); sum=0; for(i=0; i<n; i++){ scanf("%d", &A[i]); sum=sum+A[i]; } avg=(float)sum/n; printf("%f", avg); </pre>	2) <pre> int A[100][100], B[100][100], C[100][100]; int n, i, j, k; scanf("%d", &n); for(i=0; i<n; i++){ for(j=0; j<n; j++){ scanf("%d", &A[i][j]); scanf("%d", &B[i][j]); } } for(i=0; i<n; i++){ for(j=0; j<n; j++){ C[i][j]=0; for(k=0; k<n; k++){ C[i][j]=C[i][j]+A[i][k]*B[i][k]; } } } </pre>
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